

Network Management: SNMP, NETCONF, and YANG

1 Introduction

What is Network Management?

Network management includes the deployment, integration, and coordination of the hardware, software, and human elements to monitor, test, poll, configure, analyze, evaluate, and control the network and its resources to meet real-time performance and QoS requirements at a reasonable cost.

2 The Network Management Framework

A typical network management system consists of five key components:

1. Managing Server

Located in the **Network Operations Center (NOC)**, this is the central application (with humans in the loop) that controls the collection, analysis, and dispatching of management commands.

2. Managed Device

A piece of equipment on the network (router, switch, host, middlebox, etc.) with manageable hardware and software components.

3. Data (State)

Each managed device has associated "state" data:

- **Configuration Data:** Explicitly set by the manager (e.g., assigned IP address, interface speed).
- **Operational Data:** Acquired as the device operates (e.g., OSPF neighbor list).
- **Device Statistics:** Dynamic status indicators and counts (e.g., number of dropped packets, fan speed).

Note: The managing server maintains its own copy of all this data to provide a network-wide view (e.g., topology).

4. Network Management Agent

A software process running on the managed device that communicates with the server and performs local actions.

5. Network Management Protocol

Runs between the server and agent to query status and take actions. It provides the *capabilities* but doesn't manage the network itself.

3 Management Approaches

Approach 1: CLI (Command Line Interface)

Direct commands issued via console, Telnet, or **SSH**, often facilitated by scripting.

- **Pros/Cons:** Highly flexible but vendor-specific, arcane, and prone to human error.
- **Scalability:** Difficult to automate; manages devices individually. Home routers often export simple **HTTP menus** for this purpose.

Approach 2: SNMP/MIB

In use since the late 1980s. Operators query/set data in a device's **Management Information Base (MIB)**.

- **MIB Diversity:** Some are vendor-specific; others are agnostic (e.g., number of IP datagrams discarded due to header errors, or UDP segments received at a host).
- **Limitations:** While excellent for monitoring, a 2002 workshop (**RFC 3535**) identified shortcomings for configuration at scale.

Approach 3: NETCONF/YANG

The modern holistic approach to management at scale.

- **YANG:** A data modeling language used to model configuration and state data with correctness constraints.
- **NETCONF Protocol:** Used to communicate YANG-compatible actions and data remotely.
- **Key Feature:** Provides **atomic operations** across multiple devices.

NETCONF/YANG: The Modern Holistic View

This approach emphasizes configuration management with correctness constraints and **atomic operations** across multiple devices, moving away from individual device-by-device configuration.

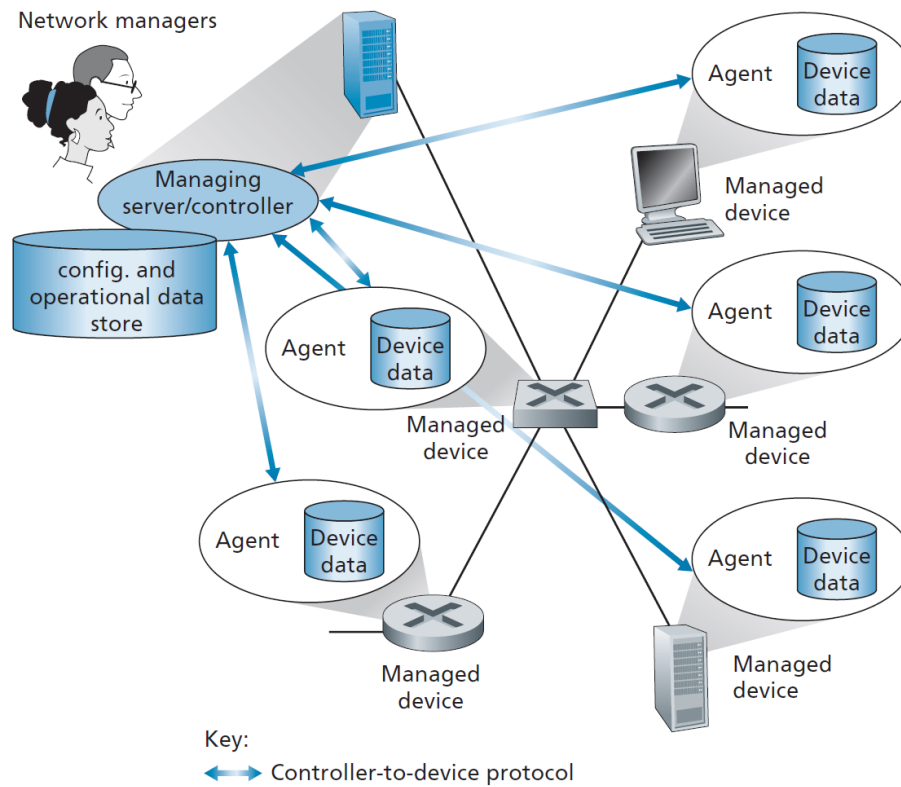


Figure 1: Elements of network management: Servers, Agents, and Managed Devices.